揚州大學

YANGZHOU UNIVERSITY

Simulation and Analysis of an Electronic Circuit Using LTspice

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1. Experiment Overview

Active filters are widely used in analog electronic systems to selectively pass desired frequency components while attenuating unwanted signals. Unlike passive filters, active filters incorporate operational amplifiers, providing higher gain, improved impedance matching, and better frequency response without requiring inductors. This experiment focuses on the simulation and analysis of an active low-pass filter implemented using the LT1022 operational amplifier in LTspice. The circuit is excited by a sinusoidal input signal while its frequency is varied over several decades. Through transient simulation, the output waveform is observed to evaluate the frequency response of the filter. The experiment demonstrates how active filters respond differently to low-frequency and high-frequency input signals and highlights the usefulness of LTspice in analyzing analog electronic circuits before practical implementation.

2. Objective

The objectives of this experiment are:

- To design and simulate an active low-pass filter using LTspice.
- To study the operating principle of an operational amplifier-based filter.
- To analyze the transient response of the circuit.
- To observe the effect of changing input frequency on the output waveform.
- To evaluate the filtering characteristics of the circuit.
- To verify the theoretical operation of an active low-pass filter through simulation.

3. Theory

An active low-pass filter is an electronic circuit that allows low-frequency signals to pass through while attenuating high-frequency components. Unlike passive RC filters, active filters utilize operational amplifiers to improve gain, provide buffering, and minimize loading effects. In this experiment, the circuit employs an LT1022 operational amplifier configured with resistors and capacitors to form a second-order active low-pass filter. The resistor-capacitor (RC) network determines the cutoff frequency, while the operational amplifier maintains signal integrity and improves overall filter performance.

The cutoff frequency of a simple RC filter is given by:

$$f_c = 1/2\pi RC$$

where

- R = Resistance (Ω)
- C = Capacitance (F)

Signals with frequencies well below the cutoff frequency experience little attenuation and appear almost unchanged at the output. As the input frequency approaches and exceeds the cutoff frequency, the capacitive reactance decreases, causing the output amplitude to decrease significantly. The LTspice simulation performs transient analysis while stepping the input frequency through multiple values (1 kHz, 10 kHz, 100 kHz, and 1 MHz). This enables direct observation of the filter's frequency-dependent behavior.

4. Software Used

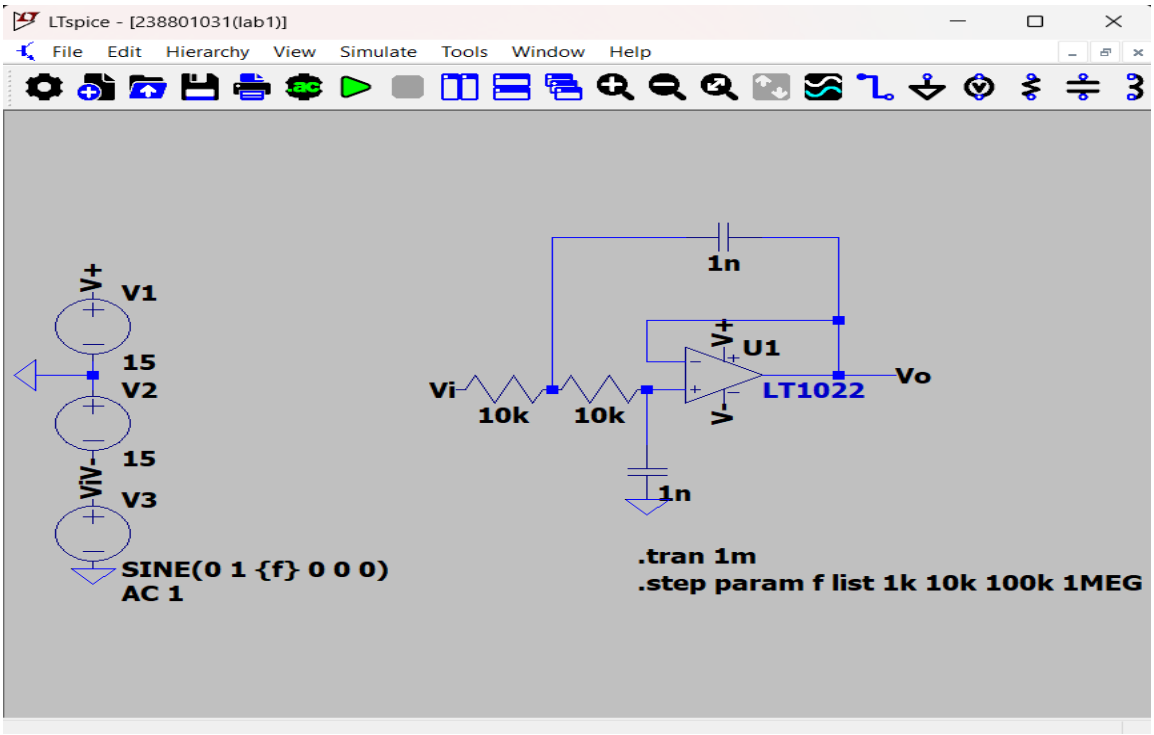
- LTspice 26.0.1
- 65 nm CMOS Technology Model
- Windows Operating System

5. Methodology

The active low-pass filter circuit was constructed using the LT1022 operational amplifier together with two resistors and two capacitors. A dual ±15 V power supply was connected to power the operational amplifier, while a sinusoidal voltage source was applied at the input terminal. Transient analysis was configured with a simulation time of 1 ms. The input signal frequency was automatically stepped through four different values (1 kHz, 10 kHz, 100 kHz, and 1 MHz) using the **.step** command in LTspice. The output voltage waveform was observed using the waveform viewer. The amplitude and shape of the output signal were compared at each frequency to evaluate the filtering performance of the circuit. The simulation results were analyzed to determine whether the circuit effectively attenuated higher-frequency signals while maintaining lower-frequency components

6. Circuit Diagram & Program:

Final outlook of the Circuit Diagram:



Coding Approach:

E:\238801031lib1.sp

*CMOS Inverter Simulation Netlist

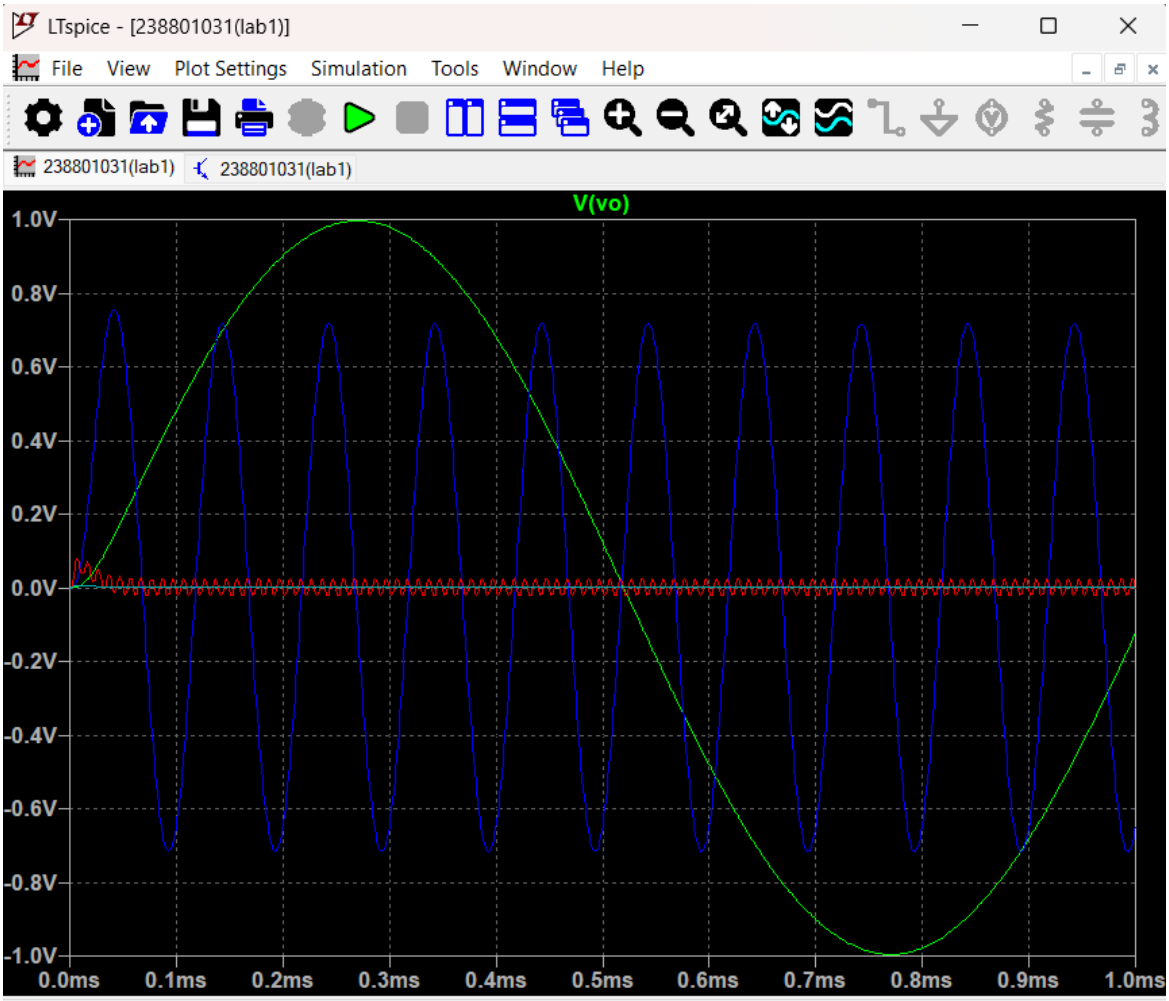
Vdd vdd 0 DC 2.5
Vin vin 0 DC 1.25

M4 vout vin vdd vdd PMOS_ MODEL W=0.75u L=0.25u
M3 vout vin 0 0 NMOS_MODEL W=0.375u L=0.25u

.model NMOS_MODEL NMOS (Vto=0.4 Kp=120u Lambda=0.05)
.model PMOS_MODEL PMOS (Vto=-0.4 Kp=40u Lambda=0.05)

.dc Vin 0 2.5 0.01
.probe
.end

Transient Simulation:



7. Discussion:

The simulation results demonstrate the expected behavior of an active low-pass filter. At low input frequencies, the capacitive reactance remains relatively high, allowing the majority of the input signal to appear at the output with minimal attenuation. As the frequency increases, the reactance of the capacitors decreases, causing a larger portion of the signal to be diverted through the RC network and reducing the output voltage. The operational amplifier provides signal buffering and maintains stable circuit performance across the tested frequency range. Compared with passive filters, the active configuration offers improved gain, higher input impedance, and lower output impedance, making it suitable for practical analog signal-processing applications.

The simulation also demonstrates the usefulness of LTspice in evaluating analog circuits before hardware implementation.

By observing the transient waveforms at different frequencies, designers can easily verify filter performance and optimize component values without constructing a physical prototype.Overall, the simulated response agrees well with the theoretical characteristics of a second-order active low-pass filter.

8. Result:

The active low-pass filter was successfully simulated using LTspice. The transient analysis confirmed that the circuit effectively passed low-frequency signals while attenuating higher-frequency components. The frequency sweep demonstrated the expected reduction in output amplitude as the input frequency increased. The simulation results closely matched the theoretical behavior of an operational amplifier-based low-pass filter.

9. Conclution:

This experiment successfully demonstrated the design and simulation of an active low-pass filter using the LT1022 operational amplifier in LTspice. The transient simulation verified the circuit's ability to selectively pass low-frequency signals while suppressing high-frequency components. The results confirmed the theoretical principles of active filter operation and illustrated the advantages of operational amplifier-based filter design, including improved signal stability, enhanced impedance characteristics, and accurate frequency response. The experiment also highlighted the effectiveness of LTspice as a powerful simulation tool for analyzing and validating analog electronic circuits before practical implementation.